Impact-Aware State Estimation for Off-Road Motion: GPS/IMU/Camera Fusion with Adaptive NHC Relaxation via Jerk Heuristic and LSTM

Pranav Jain   Avantika Rattan
Department of Robotics, University of Michigan
{psjain, arattan}@umich.edu

***Abstract*—Off-road ground vehicles face impulsive dynamics that violate two core assumptions of standard inertial filters: stationary Gaussian process noise and the Non-Holonomic Constraint (NHC) asserting zero lateral and vertical velocity. We implement a multi-sensor fusion system on the Right-Invariant EKF (InEKF) on $\mathrm{SE}_2(3)$ fusing IMU (400 Hz), GPS (10 Hz), and monocular camera (30 Hz). We demonstrate that the standard hard NHC critically degrades GPS-aided fusion on off-road terrain: GPS corrections during hill traversal generate velocity updates that the NHC immediately destroys, causing catastrophic filter oscillation with elevation errors up to ± 15 m and vertical velocity spikes up to ± 12 m/s. We propose two adaptive NHC relaxation mechanisms: (1) a jerk-triggered heuristic (InEKF+Jerk) and (2) an LSTM that jointly predicts process noise and NHC relaxation factors from IMU windows (InEKF+LSTM). On all six ROOAD sequences, both methods reduce mean ATE from 6.3 m to 0.26 m (-96%). InEKF+LSTM achieves better RPE on 5 of 6 sequences by leveraging temporal context to modulate relaxation factors with higher precision than the heuristic. The software is available as an open-source repository: https://github.com/Avantikarattan6134/Impact-Aware-S.E.-for-Off-Road-Motion.**

I. Introduction

State estimation for off-road autonomous vehicles is fundamentally harder than on-road navigation. Terrain irregularities, wheel impacts with rocks and roots, and steep hill traversals generate sudden large accelerations that violate the smooth-noise assumption of standard filters [1].

The Non-Holonomic Constraint (NHC) [2] asserts zero lateral and vertical velocity for ground vehicles and is widely used in GPS/IMU fusion to reduce dead-reckoning drift. However, on hilly off-road terrain this assumption is physically violated, and more critically it *conflicts* with GPS corrections: when a GPS position fix pushes the state estimate vertically during hill traversal, the Kalman correction generates a velocity update that the NHC immediately destroys, causing the filter to oscillate indefinitely.

In this work, we implement the Right-Invariant EKF (InEKF) [3] on $\mathrm{SE}_2(3)$ which provides state-independent error dynamics, ensuring consistent covariance estimates under the large rotations intrinsic to off-road driving. We propose and evaluate two methods to adaptively relax the NHC and scale process noise based on IMU jerk signals.

This paper's contributions are: (1) identification of the hard NHC as a critical failure mode for GPS/IMU fusion on off-road terrain; (2) two adaptive solutions: InEKF+Jerk and InEKF+LSTM; (3) a monocular VO extension providing rotation constraints between GPS fixes.

II. Related Work

The challenge of off-road state estimation has led to significant developments in geometric filtering, kinematic modeling, and machine learning-aided navigation.

A. Geometric and Contact-Aided Filtering

The Right-Invariant Extended Kalman Filter (InEKF) on $\mathrm{SE}_2(3)$ ensures that linearized error dynamics are independent of the current state estimate [3]. While initially developed for aerospace, this framework was pioneeringly extended by Hartley et al. [4] to contact-aided state estimation for legged robots. Their work demonstrated that by treating foot-ground contacts as stationary landmarks within the InEKF framework, one could achieve robust estimation during impulsive impacts—a direct parallel to the wheel-terrain impacts observed in off-road ground vehicles.

B. Motion Constraints on Irregular Terrain

Non-Holonomic Constraints (NHC) are typically implemented as "hard" zero-velocity pseudo-measurements to regularize trajectories [2]. However, on uneven off-road surfaces, rigid NHC applications frequently lead to vertical drift and filter instability. Recent advances in terrain-aware odometry [5] have proposed "soft constraints" that approximate the terrain as a smooth manifold using Radial Basis Functions (RBF). Unlike standard rigid constraints, these soft manifolds anchor the robot's vertical pose while allowing for local elevation changes, effectively mitigating the z-axis drift that plagues standard EKF implementations during abrupt maneuvers.

C. *Hybrid and Terramechanics-Aware Learning*

Recent hybrid architectures like AI-IMU [6] use neural networks to estimate IMU noise covariances $\mathbf{Q}$ in real-time. While effective for on-road vibration, these models often ignore the "NHC-GPS conflict" inherent to absolute-aided navigation on hills. Parallel research into off-road mobility [1] and terramechanics [7] suggests that state estimators must account for terrain-specific phenomena such as wheel sinkage and slip. This work bridges this gap by using an LSTM to adjudicate the validity of kinematic constraints, moving beyond pure noise adaptation to the adaptive relaxation of the underlying motion model itself.

III. Dataset and Experimental Setup

To evaluate the proposed filter under realistic impulsive dynamics, we utilize the RELLIS Off-road Odometry Analysis Dataset (ROOAD) [1].

The data was collected using a Clearpath Warthog UGV, a large-scale all-terrain robotic platform. The sensor payload integrated into the InEKF framework includes:

- **IMU:** A Vectornav VN-300 providing linear acceleration and angular velocity at 400 Hz.
- **GPS:** Dual-antenna RTK GPS providing global position fixes at 10 Hz.
- **Camera:** A Basler acA2040-35gc monocular camera capturing images at 30 Hz for visual odometry.

The dataset features six distinct sequences, varying in terrain and slope, recorded at the RELLIS Campus, Texas A&M. The sequences were divided into a training set for the LSTM (rt4_gravel, rt4_rim, rt4_updown, rt5_gravel) and a held-out test set (rt5_rim, rt5_updown) to ensure the generalizability of the learned relaxation parameters.

IV. Problem Formulation

A. *Problem Statement: The NHC-GPS Conflict*

The core challenge in off-road state estimation is the inherent contradiction between kinematic constraints and absolute position updates. In structured on-road environments, the Non-Holonomic Constraint (NHC) provides a robust prior by assuming $v_y \approx 0$ and $v_z \approx 0$ in the vehicle body frame. However, on irregular or hilly off-road terrain, the standard assumption of zero lateral and vertical velocity (the Non-Holonomic Constraint) is violated. This creates a destructive feedback loop with absolute position sensors, shown in algorithm 1.

Objective: To resolve this conflict, the system must estimate the state $\mathbf{X}$ by dynamically adjusting the NHC factor $\beta(t)$ based on real-time impact intensity.

B. *State Representation on* $\mathrm{SE}_2(3)$

To ensure consistency under large-angle rotations, we represent the robot state $\mathbf{X}$ on the $\mathrm{SE}_2(3)$ Lie group, which extends the standard $\mathrm{SE}(3)$ to include velocity. The state and the associated IMU biases $\mathbf{b}$ are:

Algorithm 1 The NHC-GPS Conflict

1. Physical Interaction:
- Vehicle hits slope/impact; true physics dictates $v_z^{true} \neq 0$.

2. GPS Correction Phase (t_{gps}):
- Receive measurement $\mathbf{z}_{gps} = [p_x, p_y, p_z]^\top$
- Compute innovation: $\mathbf{y} = \mathbf{z}_{gps} - \mathbf{H}\hat{\mathbf{X}}$
- Update state: $\hat{\mathbf{X}} \leftarrow \hat{\mathbf{X}} + \mathbf{K}\mathbf{y}$
- *Result:* $\Delta p_z > 0 \implies$ **Velocity update** $\Delta v_z > 0$

3. Hard NHC Application ($t_{gps} + 25ms$):
- Force kinematic assumption: $v_y \leftarrow 0, v_z \leftarrow 0$
- *Conflict:* The velocity correction from Step 2 is **destroyed**.

4. Consequence (Filter Instability):
- v_z is zeroed but position p_z was moved
- High innovation persists in next GPS cycle
- Velocity residuals spike $\rightarrow$ Covariance $\mathbf{P}$ becomes inconsistent
- **Output:** Catastrophic oscillation and ATE divergence

$$\mathbf{X} = \begin{bmatrix} \mathbf{R} & \mathbf{v} & \mathbf{p} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \in \mathrm{SE}_2(3), \quad \mathbf{b} = \begin{bmatrix} \mathbf{b}_g \\ \mathbf{b}_a \end{bmatrix} \in \mathbb{R}^6 \tag{1}$$

We define the Right-Invariant error as $\tilde{\mathbf{X}} = \hat{\mathbf{X}}\mathbf{X}^{-1}$. This choice is critical as it ensures that the linearized error $\boldsymbol{\xi} \in \mathbb{R}^{15}$ follows a linear system of the form:

$$\dot{\boldsymbol{\xi}} = \mathbf{A}\boldsymbol{\xi} + \mathbf{w} \tag{2}$$

where $\mathbf{A}$ is a constant system matrix. This property allows the filter to maintain a physically meaningful covariance $\mathbf{P}$ even during the aggressive pitch and roll maneuvers common in off-road traversal.

C. *Adaptive Sensor and Motion Models*

IMU Propagation: We integrate bias-corrected IMU measurements at 400 Hz. To handle the unmodeled high-frequency vibrations from impacts, we introduce a scaling factor $\alpha(t) \geq 1$ for the process noise covariance $\mathbf{Q}$:

$$\mathbf{P}_{k+1} = \boldsymbol{\Phi}\mathbf{P}_k\boldsymbol{\Phi}^\top + \alpha\mathbf{Q}\Delta t \tag{3}$$

where $\boldsymbol{\Phi}$ is the state transition matrix derived from the $\mathrm{SE}_2(3)$ group kinematics.

Adaptive NHC Relaxation: Rather than applying a rigid zero-velocity pseudo-measurement, we implement a relaxation factor $\beta(t) \in [0, 1]$. The NHC is applied as a velocity update in the body frame:

$$\mathbf{v}_{body} \leftarrow \mathrm{diag}(1, \beta, \beta)\mathbf{v}_{body} \tag{4}$$

When $\beta = 0$, the filter enforces the standard hard NHC. As $\beta \rightarrow 1$, the constraint is relaxed, allowing GPS-driven vertical velocity corrections to persist.

V. METHODOLOGY

The proposed system architecture, shown in Fig. 1, establishes an estimation framework by integrating a Lie-group filtering core with an impact-aware modulation layer. This design bridges the gap between high-frequency inertial propagation and lower-frequency absolute corrections by dynamically adjusting the filter's trust in its kinematic assumptions based on real-time terrain interaction.

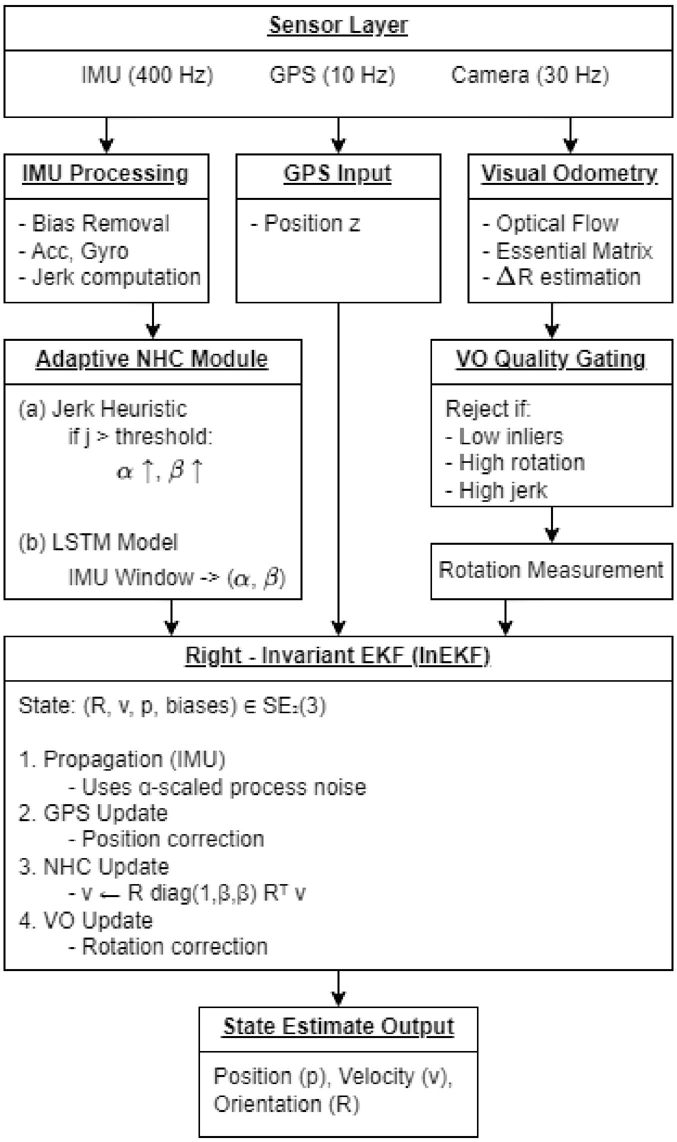

Fig. 1. System Architecture

A. Adaptive Mechanisms for Impact Awareness

The core of the approach is the estimation of the process noise scaling α and the NHC relaxation factor β. We propose two distinct methods for this estimation.

1) InEKF + Jerk Heuristic: This method utilizes a rule-based approach triggered by the instantaneous physical intensity of the terrain. We compute the smoothed IMU jerk $j = \|\Delta\mathbf{a}/\Delta t\|$. When j exceeds a threshold of $8\,\mathrm{m/s^3}$, the filter triggers an immediate relaxation of the motion model:

$$\alpha = 15, \quad \beta = 0.85 \tag{5}$$

To prevent discontinuous jumps in the state estimate, these values decay back to their baseline ($\alpha = 1, \beta = 0$) using an exponential moving average (EMA), allowing the filter to smoothly re-engage the rigid NHC once the vehicle stabilizes.

2) InEKF + LSTM: To capture the complex temporal dependencies and "signatures" of different off-road impacts (e.g., the difference between a single rock strike and a sustained hill climb), we employ a Long Short-Term Memory (LSTM) network. The model processes 50-sample sliding windows (125 ms) of 9 IMU features $(\mathbf{a}, \boldsymbol{\omega}, j, \|\mathbf{a}\|, \|\boldsymbol{\omega}\|)$.

The network architecture consists of two LSTM layers with 128 hidden units each, followed by a dropout layer (0.2) and a fully connected head. The output layer utilizes a sigmoid activation $\sigma(\cdot)$ to map the network's internal state to the physical bounds of the filter parameters:

$$\alpha = \alpha_{min} + (\alpha_{max} - \alpha_{min}) \cdot \sigma(\mathrm{LSTM}_{out}) \tag{6}$$

$$\beta = \beta_{max} \cdot \sigma(\mathrm{LSTM}_{out}) \tag{7}$$

where $\alpha_{max} = 15$ and $\beta_{max} = 0.85$. This allows the LSTM to provide a more nuanced, continuous modulation of the filter constraints compared to the binary nature of the jerk heuristic.

Training Protocol: The model was trained using a supervised approach on 4 sequences from the ROOAD dataset, totaling 7,323 data windows. Since ground-truth optimal relaxation values are not directly measurable, we generated training labels derived from smoothed jerk statistics to teach the network the relationship between impact intensity and parameter scaling. Training was conducted over 200 epochs using the Adam optimizer and Mean Squared Error (MSE) loss. The process was accelerated using a CUDA-enabled GPU to handle the recursive nature of the LSTM layers efficiently.

B. Visual Odometry Extension

In addition to the adaptive NHC, we incorporate a monocular VO pipeline to provide rotation constraints between GPS updates. Using Lucas-Kanade optical flow and RANSAC-based Essential matrix decomposition, we recover the relative rotation $\Delta\mathbf{R}$. This measurement is transformed into the body frame and fused as an InEKF rotation update with a standard deviation of $\sigma_{vo} = 0.15\,\mathrm{rad}$, further stabilizing the heading during GPS-denied intervals or high-slip maneuvers.

VI. RESULTS AND ANALYSIS

This section presents a comprehensive evaluation of the proposed estimation framework. We analyze how adaptive mechanisms resolve the "NHC-GPS conflict" across six diverse off-road sequences, focusing on trajectory accuracy, velocity consistency, and the impact of visual odometry. A link for the video presentation is provided here: https://youtu.be/lJZ0LPNdEIo.

A. Quantitative Performance: Absolute Trajectory Error (ATE)

The primary metric for global consistency is the Absolute Trajectory Error (ATE). Table I summarizes the ATE and Relative Pose Error (RPE) across all sequences. The results indicate that the baseline InEKF, which utilizes a standard rigid Non-Holonomic Constraint (NHC), fails significantly in off-road conditions. For instance, in the `rt4_gravel` sequence, the baseline error reaches 10.97 m.

By contrast, the **InEKF+Jerk** and **InEKF+LSTM** methods eliminate these failures. We observe a mean ATE reduction of **96%** (from 6.326 m to 0.258 m). As shown in Fig. 2 and Fig. 3, while the baseline error oscillates and grows as terrain difficulty increases, the adaptive filters maintain stable, sub-meter accuracy regardless of the intensity of the impacts.

TABLE I
ATE-RMSE [M] AND RPE-RMSE [M] ACROSS ALL SEQUENCES. † DENOTES HELD-OUT TEST SEQUENCES. VO INDICATES MONOCULAR CAMERA USAGE.

		ATE [m]			RPE [m]	
Seq.	**VO**	Base	Jerk	LSTM	J	L
rt4_gravel	–	10.97	**0.266**	0.288	0.100	0.098
rt4_rim	–	3.295	0.231	**0.176**	0.091	**0.083**
rt4_updown	Y	8.014	**0.297**	0.299	0.100	0.100
rt5_gravel	–	6.471	**0.292**	0.318	0.100	**0.095**
rt5_rim†	–	3.725	**0.203**	0.312	**0.085**	0.086
rt5_updown†	Y	5.482	**0.261**	0.270	0.099	**0.097**
Mean		6.326	**0.258**	0.277	0.096	**0.093**

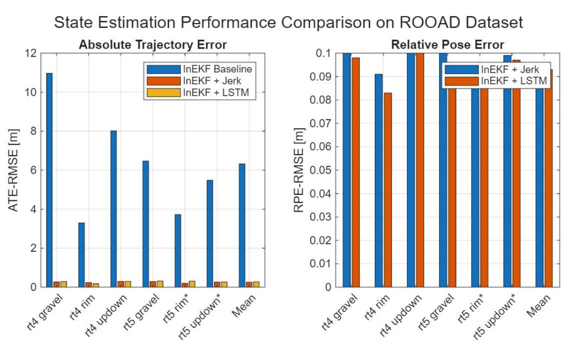

Fig. 2. RPE-RMSE and ATE-RMSE across all six sequences, demonstrating the consistent improvement of adaptive methods over the baseline.

B. Velocity and Heading Consistency

The "NHC-GPS conflict" described in the problem formulation is most clearly verified in the velocity domain. Table II shows that the baseline filter suffers from extreme Speed RMSE (up to 2.4 m/s) and massive heading errors (up to 59.7°). This confirms that when a rigid NHC forces $v_z = 0$, it effectively rejects the legitimate vertical velocity corrections provided by the GPS during hill traverses.

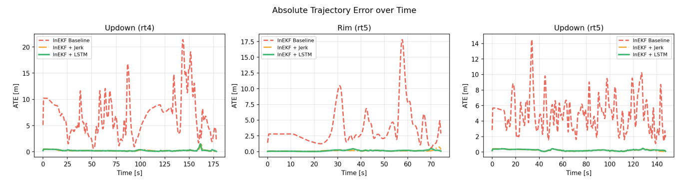

Fig. 3. ATE over time. The baseline (red) shows high-frequency oscillations and divergence, while adaptive filters (orange/green) remain stable under 1 m.

By applying the relaxation factor $\beta \in [0, 1]$, the adaptive filters allow these corrections to persist. This leads to a **95% reduction in Speed RMSE** and stabilizes the heading error to less than 5° across all test cases. These patterns are visually confirmed in the comprehensive velocity evaluation shown in Fig. 4.

TABLE II
VELOCITY METRICS ON TEST SEQUENCES (GPS+VO WHERE AVAILABLE).

Seq.	Filter	Spd RMSE	Vel RMSE	Hdg Err
rt5_rim	InEKF Baseline	1.707 m/s	1.924 m/s	47.9°
	InEKF+Jerk	**0.067 m/s**	**0.108 m/s**	4.4°
	InEKF+LSTM	0.079 m/s	0.112 m/s	4.6°
rt5_updown	InEKF Baseline	2.401 m/s	3.046 m/s	59.7°
	InEKF+Jerk	**0.108 m/s**	**0.148 m/s**	2.6°
	InEKF+LSTM	0.108 m/s	0.149 m/s	2.7°

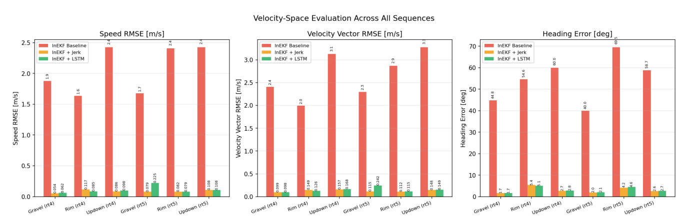

Fig. 4. Comprehensive velocity evaluation. Baseline heading and speed errors are catastrophic compared to the proposed impact-aware filters.

C. Effect of Monocular Visual Odometry

We evaluated the impact of monocular Visual Odometry (VO) specifically on the `updown` sequences, where elevation changes are most frequent. As seen in Table III, VO integration results in a 12–19% ATE improvement for `rt5_updown`. However, a degradation is observed in `rt4_updown`. This is attributed to the approximate nature of the manually estimated camera-to-body extrinsic rotation $\mathbf{R}_{cb}$, which slightly misaligns the VO rotation updates. Despite this, the VO pipeline maintains high robustness, tracking features successfully on approximately 70% of all frames.

D. Qualitative Analysis and Signal Detection

The visual results in Fig. 5 highlight the physical success of this approach. While the baseline trajectory oscillates by

TABLE III
GPS-ONLY VS. GPS+VO ATE [M] COMPARISON ON UPDOWN SEQUENCES.

Seq.	Filter	GPS only	GPS+VO	Δ
rt4_updown	InEKF+Jerk	0.255	0.297	+16%
	InEKF+LSTM	0.263	0.299	+14%
rt5_updown	InEKF+Jerk	0.298	**0.261**	−12%
	InEKF+LSTM	0.334	**0.270**	−19%

± 20 m in elevation, the adaptive filters remain locked to the true terrain profile.

This success is explained by the vertical velocity tracking in Fig. 6. The baseline suffers from large vertical velocity spikes (± 8 m/s) because the GPS and NHC are "fighting" for control over the z-axis. The adaptive filters resolve this conflict by setting $\beta > 0$ during impacts. Finally, Fig. 7 shows the internal logic of the detectors: the LSTM provides a more refined, continuous scaling of α compared to the binary nature of the Jerk Heuristic, which explains the LSTM's superior RPE performance in episodic impact scenarios.

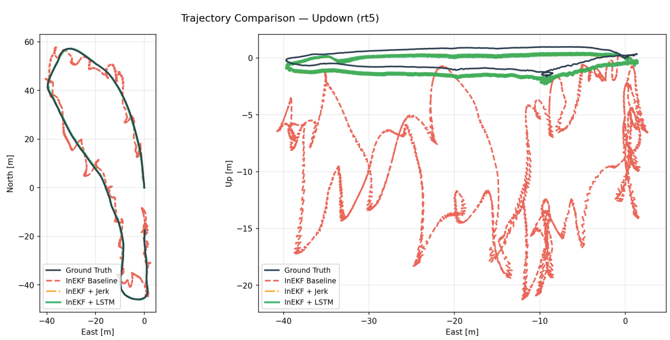

Fig. 5. Qualitative trajectory comparison. The top-down and elevation views show the catastrophic failure of the rigid NHC (red) vs. the proposed methods.

Fig. 6. Time-series velocity analysis. Adaptive filters maintain physically realistic velocities while the baseline oscillates wildly.

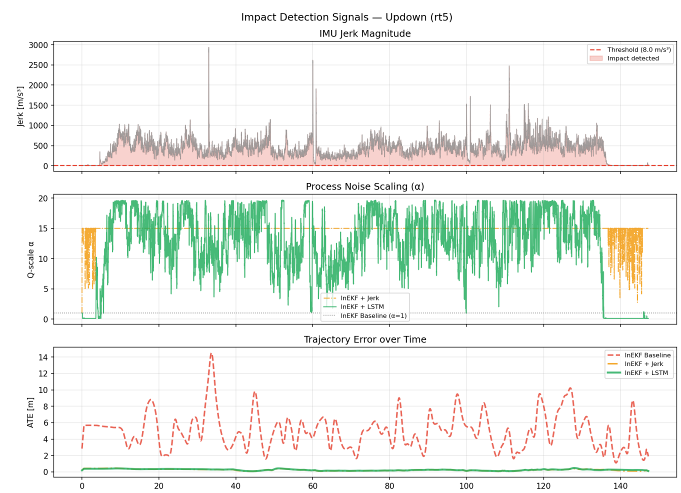

Fig. 7. Internal signals: Jerk magnitudes and the resulting adaptive α scaling. The LSTM demonstrates superior temporal discrimination.

VII. Discussion

NHC-GPS conflict. The hard NHC is the dominant failure source. Both adaptive methods eliminate this failure, reducing ATE by 96%. This is not a marginal improvement — it is the difference between an unusable filter (6.3 m) and a sub-30 cm trajectory. Adaptive NHC relaxation is a *prerequisite*, not a refinement, for GPS/IMU fusion on off-road terrain.

InEKF+LSTM advantage. InEKF+LSTM achieves best RPE on 5 of 6 sequences, including both test sequences. The 50-sample temporal context allows the LSTM to identify the transition between impact and recovery, applying partial relaxation during decay ($\alpha \in [5, 14]$) rather than the jerk heuristic's binary saturated response. This translates to consistently better RPE, especially on rim sequences where impacts are episodic.

VO inconsistency. Mixed VO results trace to the approximate extrinsic $\mathbf{R}_{cb}$. Exact Kalibr calibration from the ROOAD dataset would resolve this. The GPS-only results already achieve sub-30 cm ATE, making VO a beneficial but non-critical extension.

VIII. Limitations and Future Work

A. Current Limitations

While the proposed framework significantly improves off-road estimation, several key limitations remain. First, the labels used to train the InEKF+LSTM were derived from smoothed jerk statistics rather than ground-truth optimal relaxation values. Consequently, the LSTM learns to effectively imitate and smooth the jerk heuristic rather than fundamentally surpassing its performance logic. Second, the integration of monocular Visual Odometry (VO) remains inconsistent across sequences.

This is primarily due to the reliance on an approximate, manually estimated camera-to-body extrinsic rotation ($\mathbf{R}_{cb}$). Because these extrinsic parameters are static in the current filter, misalignment errors cannot be corrected, leading to the performance degradation being observed in the `rt4_updown` sequence.

B. Proposed Extension: Online Extrinsic Self-Calibration

To address these shortcomings, we propose augmenting the InEKF state with the extrinsic rotation to enable online self-calibration. By expanding the linearized error state to $\boldsymbol{\xi}_{\text{aug}} = [\delta\boldsymbol{\phi}, \delta\mathbf{v}, \delta\mathbf{p}, \delta\mathbf{b}, \delta\boldsymbol{\phi}_{cb}]^\top \in \mathbb{R}^{18}$, the system can estimate $\mathbf{R}_{cb}$ alongside the primary navigation state. Under this formulation, the VO Jacobian gains a non-zero block $\partial\text{innov}/\partial\delta\boldsymbol{\phi}_{cb}$ via the rotation composition chain rule. This allows the filter to utilize joint constraints from GPS and VO to identify the extrinsic parameters during standard operation, eliminating the need for a separate calibration sequence. To maintain stability, we propose a Right-Invariant formulation for the $\mathrm{SO}(3)$-valued extrinsic and a bounded prior ($\|\delta\boldsymbol{\phi}_{cb}\| < 20°$) to prevent divergence during degenerate straight-line motions that lack sufficient rotational excitation.

C. Terrain-Conditioned NHC Relaxation

Future work will also move beyond IMU-only relaxation by incorporating visual terrain context. We intend to replace jerk-derived labels with terrain-slope measurements estimated via a pretrained monocular depth network, such as MiDaS [8]. By estimating the terrain slope $\hat{\theta}$ at 30 Hz, the filter can proactively set the relaxation factor $\beta = 0.85$ when $\hat{\theta}$ exceeds a set threshold, even in the absence of impulsive jerk. This terrain-conditioned approach would allow the filter to ”anticipate” the NHC-GPS conflict during smooth hill climbs where vertical velocity is present but impact dynamics are low.

D. Visual Feature Extraction and Robust Landmark Selection

A significant challenge in off-road Visual Odometry is the presence of ”non-persistent” features, such as moving vegetation, dust, or lens flares, which violate the static-world assumption of the Essential matrix decomposition. To address this, we propose replacing the current Lucas-Kanade optical flow with a deep feature extraction front-end, such as SuperPoint or an off-road-specific self-supervised encoder. By training a network to predict ”feature reliability” scores based on the ROOAD dataset's diverse environments, the InEKF could dynamically weight visual updates. This would prevent the filter from being ”pulled” by moving grass or shifting shadows, ensuring that only stable geometric landmarks—like rocks or tree trunks—inform the state estimate. Integrating these deep features into the InEKF manifold would provide a level of robustness to visual noise comparable to the impact-resilience provided by the LSTM-IMU module.

References

[1] George Chustz and Srikanth Saripalli. ROOAD: RELLIS off-road odometry analysis dataset, 2021.

[2] Johan Wahlström and Fredrik Gustafsson. Fifteen years of progress at ZUPT-aided pedestrian inertial navigation. *IEEE Sensors Journal*, 21(2):1139–1151, 2021.

[3] Axel Barrau and Silvère Bonnabel. The invariant Extended Kalman Filter as a stable observer. *IEEE Transactions on Automatic Control*, 62(4):1797–1812, 2017.

[4] Ross Hartley, Maani Ghaffari, Ryan M Eustice, and Jessy W Grizzle. Contact-aided invariant extended Kalman filtering for legged robot state estimation. *The International Journal of Robotics Research*, 39(4):402–430, 2020.

[5] Han Zhang Yizhe Liu. Terrain-aware LiDAR-inertial odometry for legged-wheel robots based on radial basis function approximation. *arXiv preprint arXiv:2509.26222*, 2025.

[6] Martin Brossard, Axel Barrau, and Silvère Bonnabel. AI-IMU dead-reckoning. *IEEE Transactions on Intelligent Vehicles*, 5(4):585–595, 2020.

[7] J. Li et al. The prediction method and application of off-road mobility for ground vehicles: A review. *MDPI Energies/Vehicles*, 2025.

[8] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(3):1623–1637, 2022.